Sahaj Bhadja

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Summary

Design Engineer with 2+ years of experience building **React/TypeScript interfaces**, **interactive prototypes**, AI-enabled experiences, APIs, and **cross-platform applications** for responsive user workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **responsive data interfaces**, SQL indexing, and query optimization, supporting 500+ users with fast, maintainable analytics workflows across tenant environments.
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable internal software workflows and technical operations.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production user platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **maintainable service components**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Increased lead conversions by 15% through **customer and lead-management experiences**, integrating role-based access, WebSocket updates, notifications, Azure SQL, and modular services across eight operational product domains reliably.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms interactive workflows** across 50K+ booking records, integrating pricing, coupons, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.

Skills

Languages: TypeScript, JavaScript, HTML, CSS, Python, Java, SQL
Frontend & Prototyping: React, React Native, Next.js, Angular, Responsive Design, Reusable Components, UI Development, Cross-Platform Development, Dashboard Development
APIs & Data: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, JSON, PostgreSQL, MongoDB, Firebase, Azure SQL
AI & Media: Transformers, PyTorch, LLMs, BLIP, GIT Captioning Model, ViT-GPT-2, Image Captioning, Image Segmentation, Stable Diffusion, CLIP, Content-Based Search
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Vercel, Cloud Storage
Engineering: Object-Oriented Programming, System Design, Unit Testing, Integration Testing, Automated Testing, Debugging, Monitoring, Error Handling, Agile, Scrum, Technical Documentation

Projects

Interactive Streaming Experience - Design Concept
- Designed an **interactive streaming prototype** concept with responsive React interfaces, simulated playback data, personalized content rails, keyboard and remote navigation, reusable components, experimentation controls, and stakeholder-ready demonstration flows.

CaptionCraft | Live
- Benchmarked BLIP, GIT, and ViT-GPT-2 on 10K images and 100K captions, improving **AI-generated caption accuracy** from 81% to 96% while building scalable real-time inference interfaces for media-rich user experiences.

PiXMod | Live
- Built an **AI-powered visual editing experience** with Next.js, MongoDB, segmentation, generative restoration, content-based search, responsive workflows, and subscription features supporting interactive image manipulation across user sessions reliably.